

Conditional diffusion emulator of Veros ACC equilibrium states: results

5 October 2026

Summary

- Goal: generate the post-spin-up (T, S) state of the Veros ACC channel from the EKE-closure coefficients (c_k, c_ϵ) , skipping the spin-up.
- Pipeline: DINO-Fusion diffusion code adapted to a parameter ensemble, with a velocity target, an unclipped sampler and the loss restricted to the water cells; one split runs end to end in under one A100 hour. A first round with DINO-Fusion's ϵ -prediction and clipped sampler gave 0.06–0.15 K on the same splits; the present configuration is better on every metric and is used throughout.
- Interpolation (scattered points, a band of three rows, the 5×5 and 7×7 centre blocks held out): hold-out RMSE 0.029 / 0.048 / 0.037 / 0.041 K against 0.037 / 0.116 / 0.077 / 0.114 K for the nearest training run and 0.155–0.197 K for the mean state. The model's error does not depend on the size of the gap; the baselines' does, so the model wins as soon as the gap exceeds one grid step.
- Extrapolation (the 51 or 75 outer runs held out, training on the centre; the top row): 0.079 / 0.123 / 0.074 K against 0.131 / 0.155 / 0.170 K, medians 0.041 / 0.049 / 0.054. The mean is set by the warm corner, where the model is cold-biased (corner run 0.61, 1.05 and 0.21 K when it lies two, three and one cells from the training set); elsewhere 0.03–0.15 K.
- Four training runs only, one step in from each corner: 0.150 vs 0.179 K for the nearest run and W_1 0.090 vs 0.156 K; the model interpolates the warm-side trend the baseline cannot, and loses where the state hardly changes.
- Samples: single-sample RMSE 0.07–0.11 K in interpolation, ensemble spread 0.05–0.07 K against a true within-window 0.03 K; S within 0.05 psu of its constant; land exact.
- Physical miss: temperature inversions are generated at the rate of the weakly eddying runs in every regime.

1 Data

- 100 Veros ACC runs on a 10×10 log-spaced grid, $c_k \in [0.0125, 0.8]$, $c_\epsilon \in [0.0875, 5.6]$; 100 years, 30-day output; interior grid $30 \times 42 \times 15$ (x, y, z).
- Kept: T and S , last 20 years (241 snapshots per run). $S = 35$ psu everywhere in every run.
- Nearly deterministic end states: within-window std 0.01 K above 1400 m vs 0.05 K between grid neighbours; the bottom level still drifts by 0.45 K over the window.
- One effective parameter: $\log c_k - 1.05 \log c_\epsilon$ explains 99 % of the variance of the domain-mean T ; the low- c_k/c_ϵ part of the grid saturates.
- Per-level distribution of T (Figure 1): above 400 m a mixture of restoring values and interior, with σ_z set by the meridional gradient, so $\mu_z \pm 3\sigma_z$ is wider than the level and the normalised values stay within $[-0.85, 0.6]$; below 1400 m a warm tail to $x' = +2.7$. Run-to-run signal σ_z^{runs} : 0.7 % of $3\sigma_z$ at the surface, 26 % at the bottom. The narrow peaks are the zonally uniform southern rows $y = 0-6$, the restoring zone: one peak per row, at the same value in every run to within 0.04–0.18 K (Figure 2).

Distribution of T per level, all 100 runs $\times$ 241 snapshots, water cells (surface at the top). Black: μ_z ; red dashed: $\mu_z \pm 3\sigma_z$; orange ticks: the 100 per-run level means.

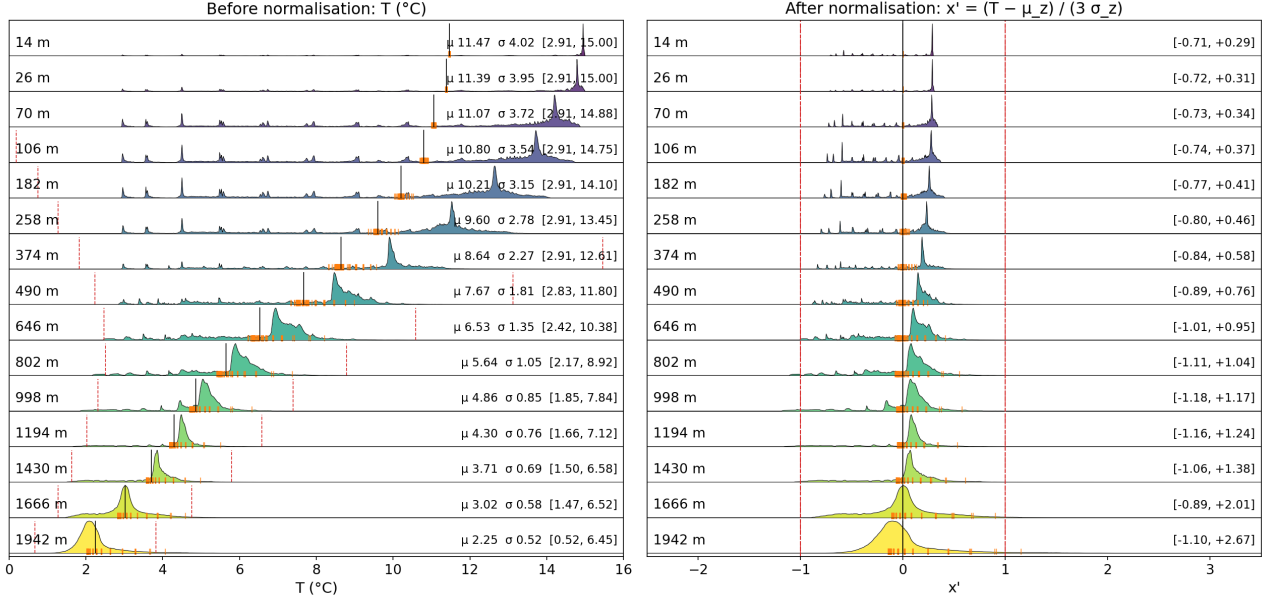


Figure 1: Distribution of T per level over all 100 runs and 241 snapshots (water cells), before (left) and after (right) the per-level normalisation of Eq. (1). Black: μ_z ; red dashed: $\mu_z \pm 3\sigma_z$; orange ticks: the 100 per-run level means.

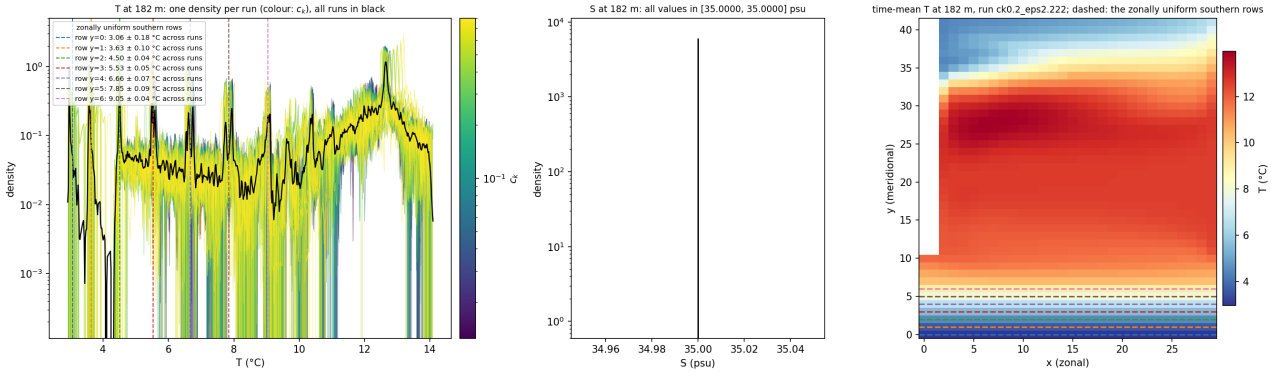


Figure 2: The level nearest 150 m (182m), last 20 years, water cells. Left: one density of T per run (colour: c_k ; log scale), all runs in black; dashed: the zonal-mean T of the southern rows $y = 0-6$, each with its spread across runs. Middle: S , exactly 35 psu everywhere. Right: time-mean T of one run at that level with the same rows marked. Every narrow peak of the distribution is one southern row.

2 Method

- Fields $\rightarrow$ 30 channels ($T_{1..15}, S_{1..15}$), normalised per level, land $\rightarrow$ 0, zero-padded $42 \times 30 \rightarrow 48 \times 32$:

$$x'_z = \frac{x_z - \mu_z}{3 \max(\sigma_z, 0.05)}. \quad (1)$$

μ_z, σ_z are computed once on all 100 runs and held fixed for every split (a deliberate, mild leakage of 30 scaling constants; no held-out state enters training).

- Denoiser: UNet2DModel (64, 64, 128, 128), 6.9 M parameters. The condition c , the standardised pair $(\log c_k, \log c_e)$, enters through an MLP added to the timestep embedding.
- DDPM with the velocity target, 1000 steps, squared-cosine schedule, EMA; the loss runs over the water cells only:

$$\mathcal{L}(\theta) = \mathbb{E}_{t, x_0, \epsilon} \|v_t - v_\theta(x_t, t, c)\|_{\text{water}}^2, \quad x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon, \quad v_t = \sqrt{\alpha_t} \epsilon - \sqrt{1 - \alpha_t} x_0. \quad (2)$$

20 000 steps, batch 32, AdamW 10^{-4} with cosine decay; 17–25 min on one A100.

- Sampling: 1000-step DDPM loop with EMA weights; the clean-state estimate $\hat{x}_0 = \sqrt{\bar{\alpha}_t} x_t - \sqrt{1 - \bar{\alpha}_t} \hat{v}$ is used unclipped; after every step the land and padding cells are reset to zero at the step’s noise level, $\sqrt{1 - \bar{\alpha}_t} z$, as in training. 32 samples per condition.
- Why the velocity target and the water-only loss. With ϵ -prediction the clean-state estimate divides by $\sqrt{\bar{\alpha}_t}$, so the chain needs a clip on $\hat{x}_0$, and the clip at $|x'| \leq 1$ caps the deep levels, whose data reach +2.7; the velocity form stays bounded and runs unclipped. Its target, however, has a step at the land and padding border (noise only there, noise minus the clean state in the water), and with the loss on all cells the network built that step with a localised activation spike in its decoder: a 3×3 patch of wrong values, 0.3–0.9 K through the water column, at a seed-dependent place next to a zero border, invisible to the loss. Restricting the loss to the water cells removes the step and the spike (Figure 19); the evaluation summary flags any such patch (worst column of the single-sample error and the zonal correlation of its error).

Differences from the DINO-Fusion code.

- Conditioning: DINO’s UNet is unconditional and is steered at sampling time by constraints; here the UNet receives $(\log c_k, \log c_e)$ through the timestep embedding, and the data are an ensemble of 100 runs split by run instead of one trajectory.
- Sampling: DINO’s BorderZeroConstraint resets land and padding to exact zeros after every step, which the network never sees in training, where those cells carry noise like every other. Here they are reset to zeros at the step’s noise level, $\sqrt{1 - \bar{\alpha}_t} z$, exactly as in training. DINO’s stratification constraints are not ported yet.
- Training target: velocity instead of noise, loss on the water cells only, no clip at sampling.
- Everything else is kept: transform chain, 3-std normalisation, DDPM settings, EMA.

3 Evaluation

- Truth: 20-year time-mean state of each held-out run; metrics on water cells.
- Baselines: mean training state; nearest training run in standardised log-parameter space; average of the axis-neighbour training runs on the grid.
- Eight hold-out splits, same data file and statistics, different training sets. Interpolation: scattered (10 interior points), band of three rows ($c_k = 0.126, 0.2, 0.3175$; 30 runs; training rows a factor 6.3 apart), centre 5×5 (c_k 0.05–0.32, c_e 0.35–2.2; 25 runs) and centre 7×7 (c_k 0.0315–0.504, c_e 0.22–3.5; 49 runs; training on the outer rows and columns only). Extrapolation: outer 51 (the complement of the 7×7 block, training on its centre), outer 75 (the complement of the 5×5 block, training on 25 runs), top row ($c_k = 0.8$; 10 runs). Four runs: training on $c_k \in \{0.0198, 0.504\} \times c_e \in \{0.139, 3.53\}$, one step in from each corner; 96 held out.
- Distributional metric: each state is reduced to its profile of horizontal means over water cells, $\bar{T}_z$; per level, the Wasserstein-1 distance between the 32 generated values and the 241 true snapshots of the window, through the quantile functions,

$$W_1^{(z)} = \int_0^1 |F_{g,z}^{-1}(q) - F_{t,z}^{-1}(q)| dq, \quad W_1 = \sum_z \frac{\Delta z_z}{H} W_1^{(z)}, \quad (3)$$

with Δz_z the level thicknesses ($H = 2080$ m). All 241 snapshots are used: the quantile form accepts unequal sizes and a subsample of 32 would only add noise. A point prediction x (the baselines) gives $W_1^{(z)} = \frac{1}{241} \sum_i |x_z - \bar{T}_{z,i}|$. W_1 rewards a correct spread and does not reward collapse to the mean; the RMSE keeps the horizontal structure.

split	runs	RMSE (K)				median (K)	bias (K)	W_1 (K)			inversions (%) generated / truth
		mean of 32	sample	nearest	mean state			diffusion	nearest	mean state	
scattered	10	0.029	0.065	0.037	0.155	0.027	−0.00	0.029	0.044	0.136	9.1 / 7.7
band (three rows)	30	0.048	0.078	0.116	0.197	0.048	+0.00	0.037	0.095	0.160	10.0 / 1.3
centre 5×5	25	0.037	0.070	0.077	0.157	0.034	+0.01	0.031	0.069	0.133	9.2 / 3.6
centre 7×7	49	0.041	0.074	0.114	0.197	0.038	+0.00	0.032	0.095	0.166	9.5 / 4.2
outer 51	51	0.079	0.110	0.131	0.258	0.041	−0.02	0.051	0.117	0.223	10.6 / 6.6
outer 75	75	0.123	0.149	0.155	0.219	0.049	−0.07	0.094	0.137	0.189	10.4 / 6.1
top row	10	0.074	0.108	0.170	0.470	0.054	+0.01	0.049	0.142	0.420	11.6 / 0.1
four runs	96	0.150	0.168	0.179	0.254	0.107	+0.01	0.090	0.156	0.208	10.5 / 5.4

Table 1: Hold-out metrics per split, mean over the held-out runs, water cells, 32 samples per run. Outer 51 and outer 75: the complements of the 7×7 and 5×5 blocks, training on the block; four runs: training on one run one step in from each corner. Bias: domain mean of the ensemble mean. The neighbour-average baseline equals the nearest run except in the scattered split (RMSE 0.020 K, W_1 0.036 K); the ensemble spread is 0.05–0.07 K against a true within-window spread of 0.03 K.

4 Findings

- **Interpolation: the model’s error is gap-independent, the baselines’ is not.** Band: 0.048 against 0.116 K for the nearest run, 22 wins of 30, better at every level below 14 m. Centre 5×5 and 7×7 : 0.037 and 0.041 K, flat over the block (0.02–0.08 K), while the nearest run goes from 0.077 to 0.114 K and reaches 0.19 K at the centre of either block, where the model stays at 0.03 K; 18 of 25 and 35 of 49 wins. The losses are the runs of the flat regime (low c_k , high c_e), where a copied neighbour is already within 0.03 K; in the scattered split, whose neighbours are all one step away, the two tie (0.029 vs 0.037 K, 5 wins of 10) (Figures 3–6).
- **Extrapolation: the warm corner sets the mean, the rest behaves like interpolation.** Outer 51 / outer 75 / top row: mean 0.079 / 0.123 / 0.074 K, median 0.041 / 0.049 / 0.054 K. The corner run ($c_k = 0.8$, $c_e = 0.0875$) errs by 0.61, 1.05 and 0.21 K with a cold bias (−0.28, −0.76, 0.05 K domain mean): the model does not carry the warming beyond the last training row, and the error grows with the distance to it (two, three and one cells). Away from the corner the outer 75 sit at 0.03–0.15 K. Wins against the nearest run: 30 / 51, 48 / 75 and 9 / 10; the nearest run is worse still at the corner (0.92, 1.24 and 0.35 K) (Figures 7–9).
- **Four training runs.** Mean 0.150 vs 0.179 K for the nearest run and 0.254 K for the mean state, W_1 0.090 vs 0.156 K, but median 0.107 vs 0.070 K and 32 wins of 96. The model produces a smooth field over the grid, 0.03–0.13 K in the flat regime and 0.15–0.5 K on the warm side, 0.51 K at the corner (cold); the nearest run is better where the state hardly changes (0.01–0.07 K) and worse where it does (0.3–0.85 K in the warm half of the interior): inside the training rectangle 0.157 vs 0.191 K, outside 0.141 vs 0.165 K. Below 250 m the model wins at every level (Figure 10).
- **Depth structure.** 0.01–0.06 K at every level in the interpolation splits, with the bottom two levels at 0.02–0.05 K; the warm corner lifts the extrapolation splits to 0.09–0.18 K between the surface and 500 m, where its front lives. The surface is the worst level in kelvin for the distribution: W_1 on the horizontal-mean profile is 0.065–0.08 K at 14 m against a cell RMSE of 0.035–0.05 K for the ensemble mean, so each sample carries a nearly uniform offset of the surface mean (0.005 of the 12 K scale) that the average of 32 removes.
- **The decoder spike and its fix.** With the loss on all cells, every velocity-target training grew one 3×3 patch of wrong values next to a zero border, in temperature and salinity alike, the same at every condition and unchanged by any sampler variant. Probing the network showed a flat encoder and one cell of the second decoder block at $2.7\times$ the median activation, amplified to $15\times$ at full resolution; true snapshots noised to a mid noise level and denoised in one step came back with a $4\times$ larger error in the patch (Figure 19). With the loss on the water cells the maps are flat in the water, the one-step error is uniform, and the eight models above show no patch. The top-row model’s largest columns (0.3–0.4 K, S off by 0.05 psu) sit in one cell at the north-east corner of its warm runs, where the front meets the northern boundary; a second seed reproduces them at the same place with the same error (0.077 vs 0.074 K overall), so this is a boundary-cell error of the

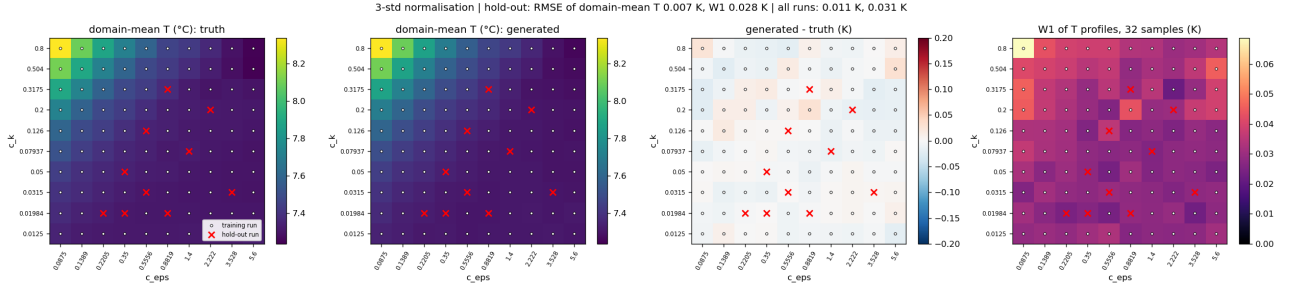
extrapolated regime, not a seed-dependent spike.

- **Inversions.** 9–12 % of interfaces with $\partial_z T < 0$ in every regime; truth 7.7 % (scattered), 6.6 and 6.1 % (outer 51 and 75), 4.2 and 3.6 % (centre 7×7 and 5×5), 1.3 % (band), 0.1 % (top). Not learned.
- **W_1 agrees with the RMSE ranking.** Training grid points score like held-out ones (0.028–0.033 vs 0.028–0.037 K in the interpolation splits); from 250 m down the samples beat the point baselines in every split, which pay for the spread they cannot represent.
- **Plumbing.** S within 0.01–0.05 psu of 35 with no constraint; land exact; ensemble spread 0.05–0.07 K against a true within-window 0.03 K (Figures 11–18).

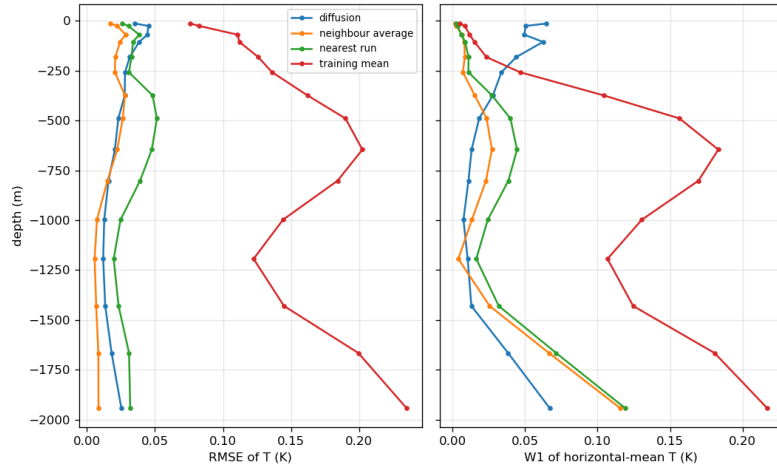
5 Next steps

- Stratification constraint at sampling time (DINO’s isotonic projection on T): targets the inversions.
- Warm corner: a limit of the extrapolation, not of the sampler; stronger conditioning or an explicit trend prior is the lever.
- Stronger conditioning: classifier-free guidance; the year as a third condition (bottom drift of 0.45 K inside the window).
- Evaluate against snapshots as well as the time mean (nearest-snapshot RMSE, spread–skill).
- Then the same pipeline on a global Veros configuration (active S , seasonal cycle, multi-century spin-up).

Reproducibility. Branch `miniveros` of the DINO-Fusion repository, package `miniveros/`; configuration `prediction_type=v_prediction, clip_sample=false, mask_loss=true`. Training at `3155a6b` (centre 5×5 , outer 75), `a94136b` (four runs) and `dd05902` (the others), with identical training code; sampling and evaluation at `004eb36` or later; the spike diagnostics (`diag_spot.py`, `diag_act.py`) at `c59f936`. Extraction 2.5 min (8 cores), training 17–25 min (one A100), generation of the hold-out and grid sets plus evaluation 15–35 min.

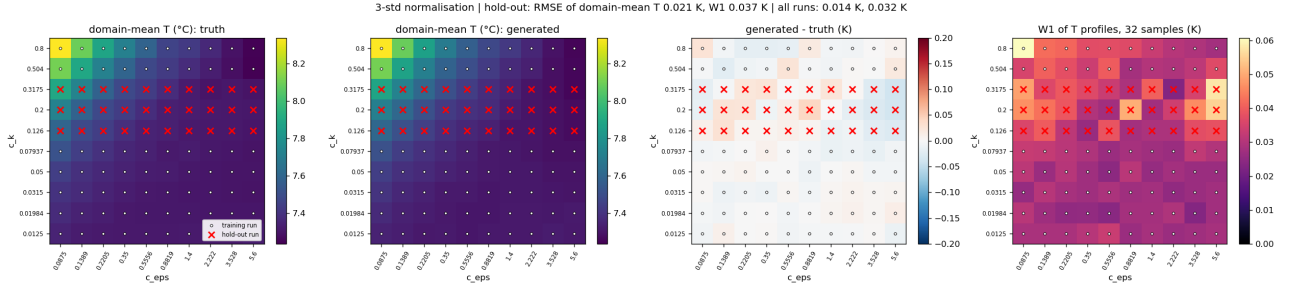


(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.

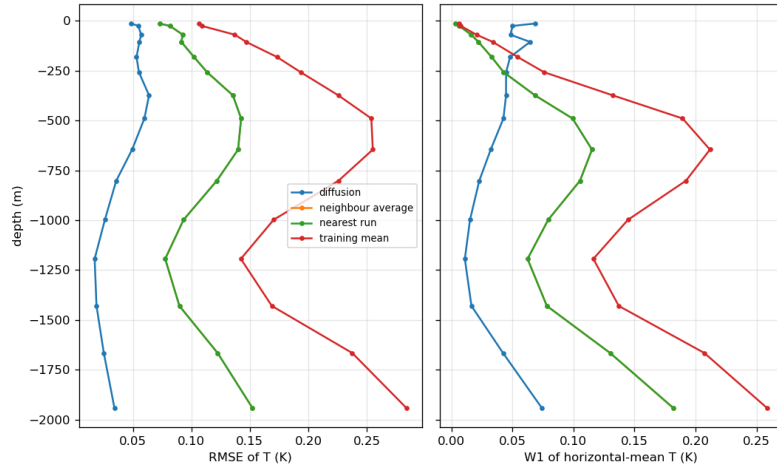


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 3: Scattered split (10 interior held-out runs), 32 samples per run.

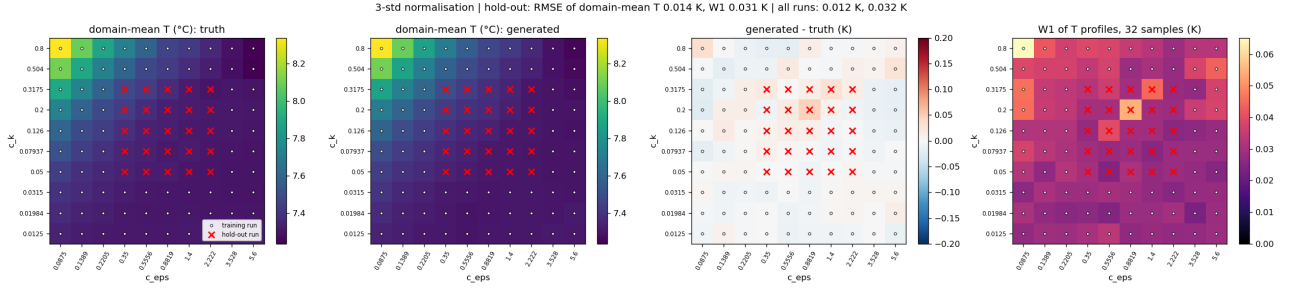


(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.

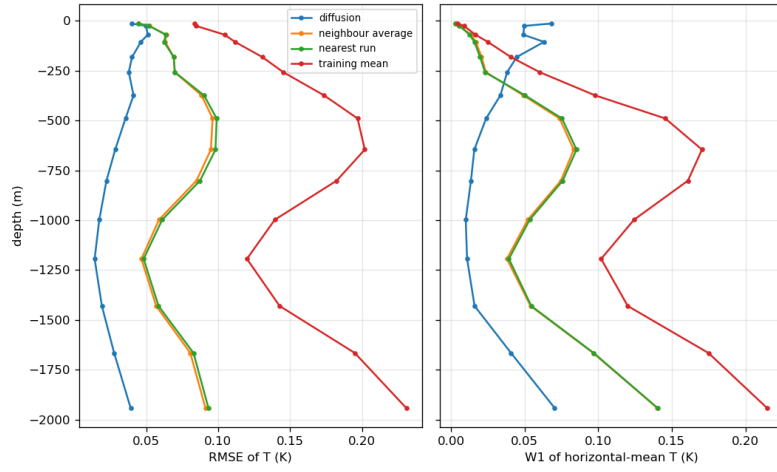


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 4: Band of three rows ($c_k = 0.126, 0.2, 0.3175$ held out), 32 samples per run.

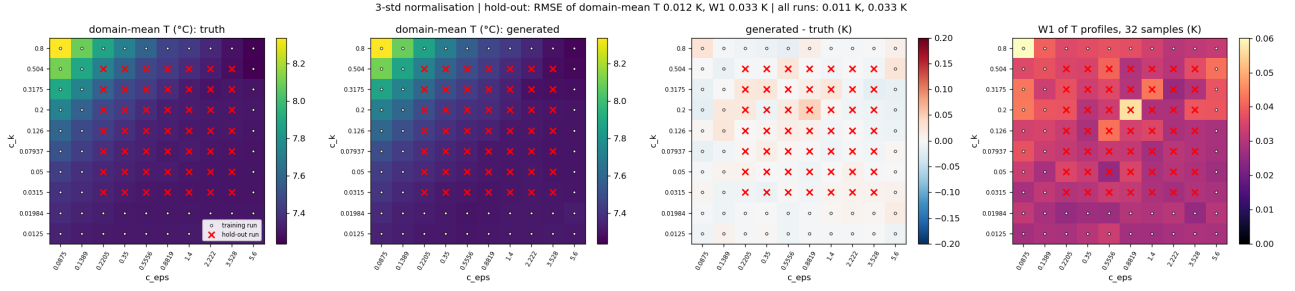


(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.

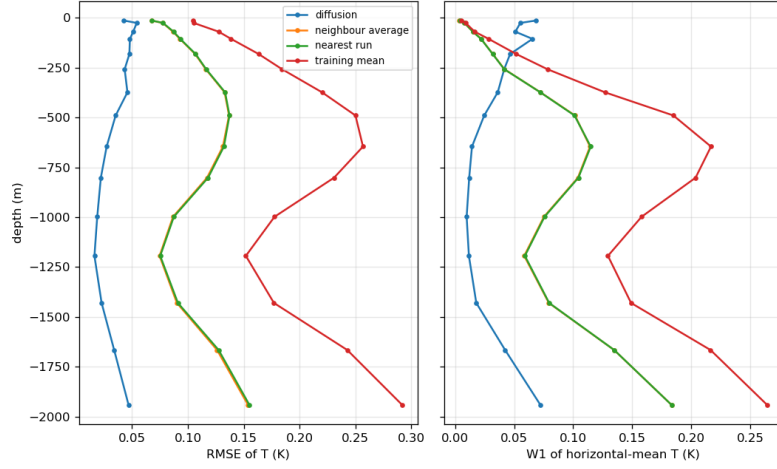


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 5: Centre 5×5 block held out (25 runs; training on the 75 outer runs), 32 samples per run.

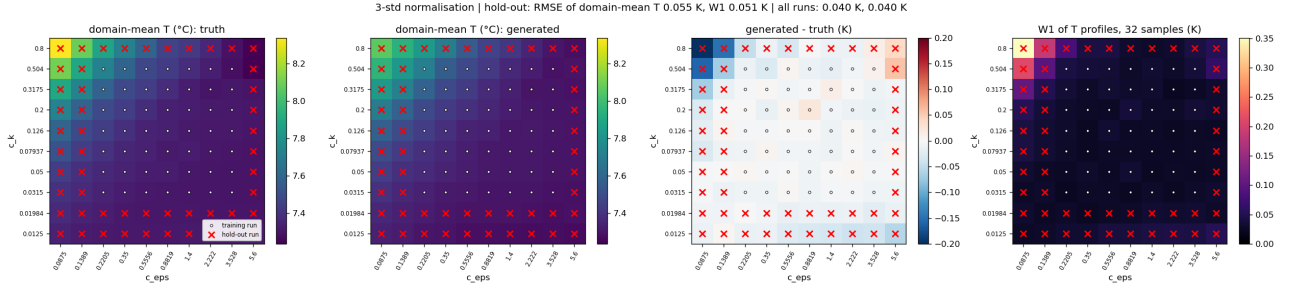


(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.

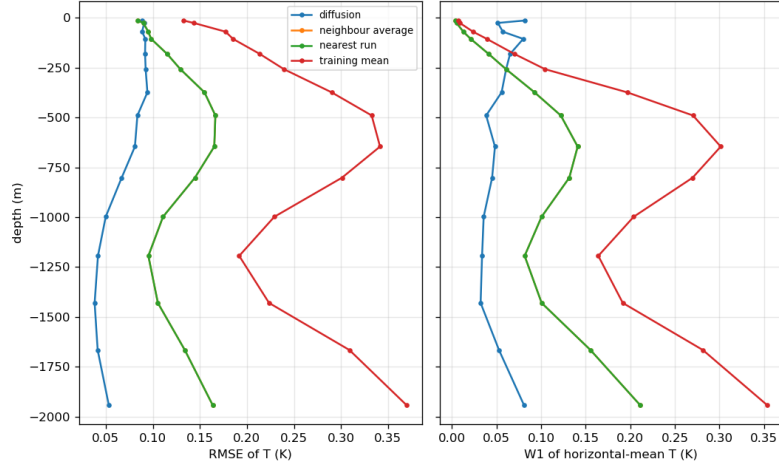


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 6: Centre 7×7 block held out (49 runs; training on the 51 edge runs), 32 samples per run.

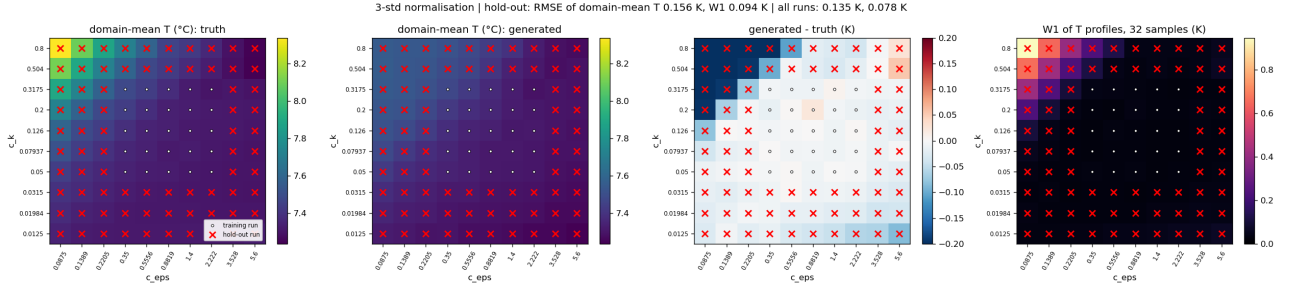


(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.

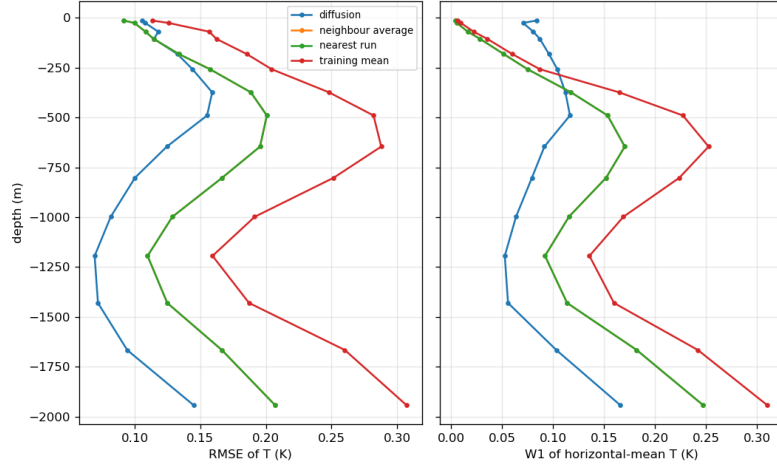


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 7: Outer 51 held out (training on the 7×7 centre), 32 samples per run.

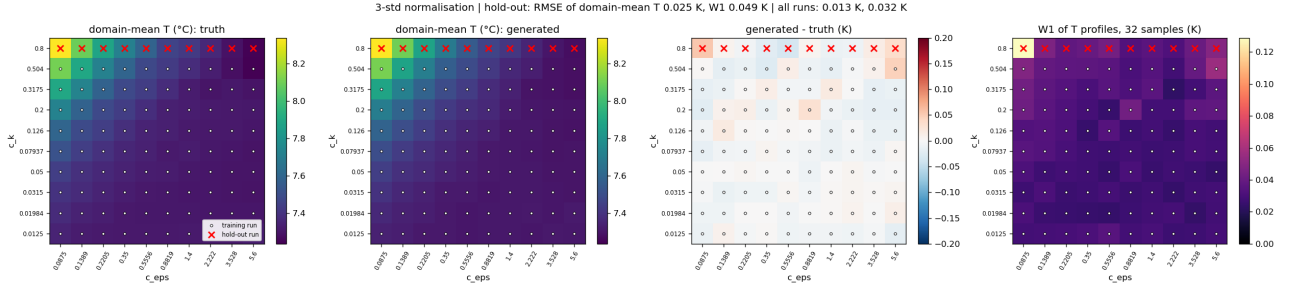


(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.

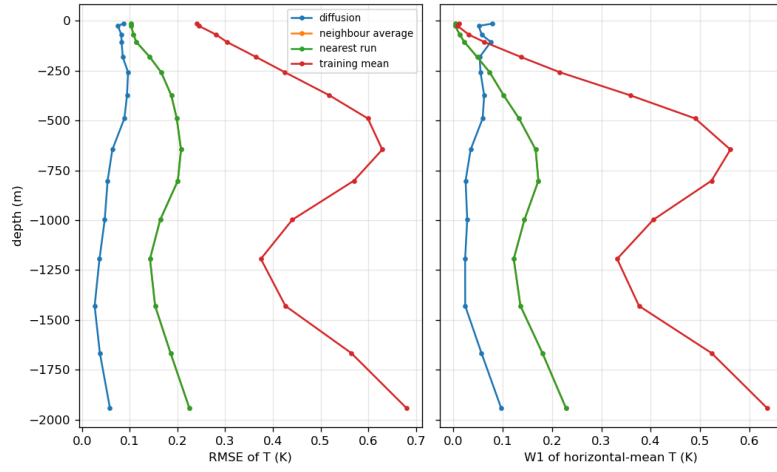


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 8: Outer 75 held out (training on the 5×5 centre), 32 samples per run.

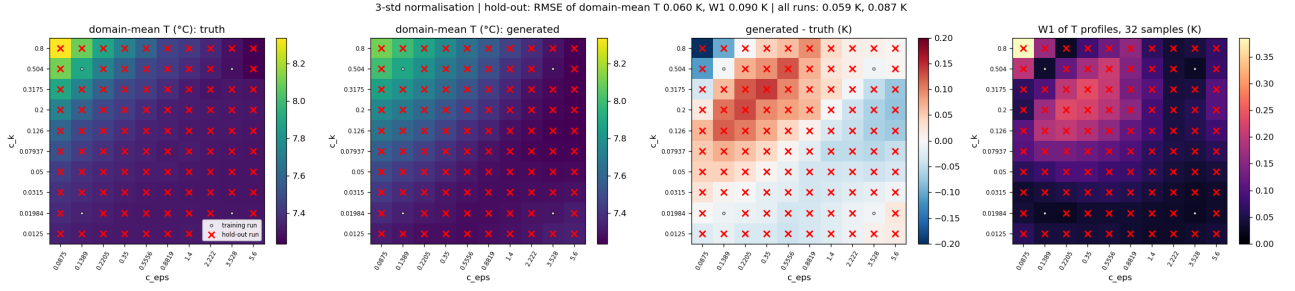


(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.

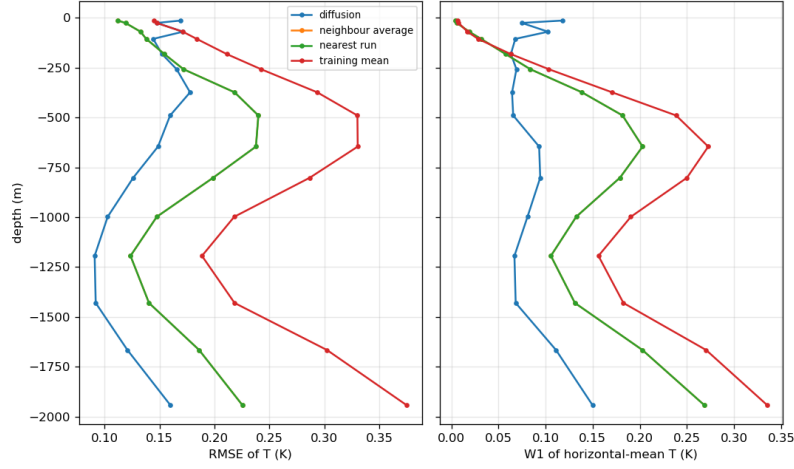


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 9: Top row held out ($c_k = 0.8$, 10 runs), 32 samples per run.



(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.



(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 10: Four training runs, one step in from each corner (96 held out), 32 samples per run.

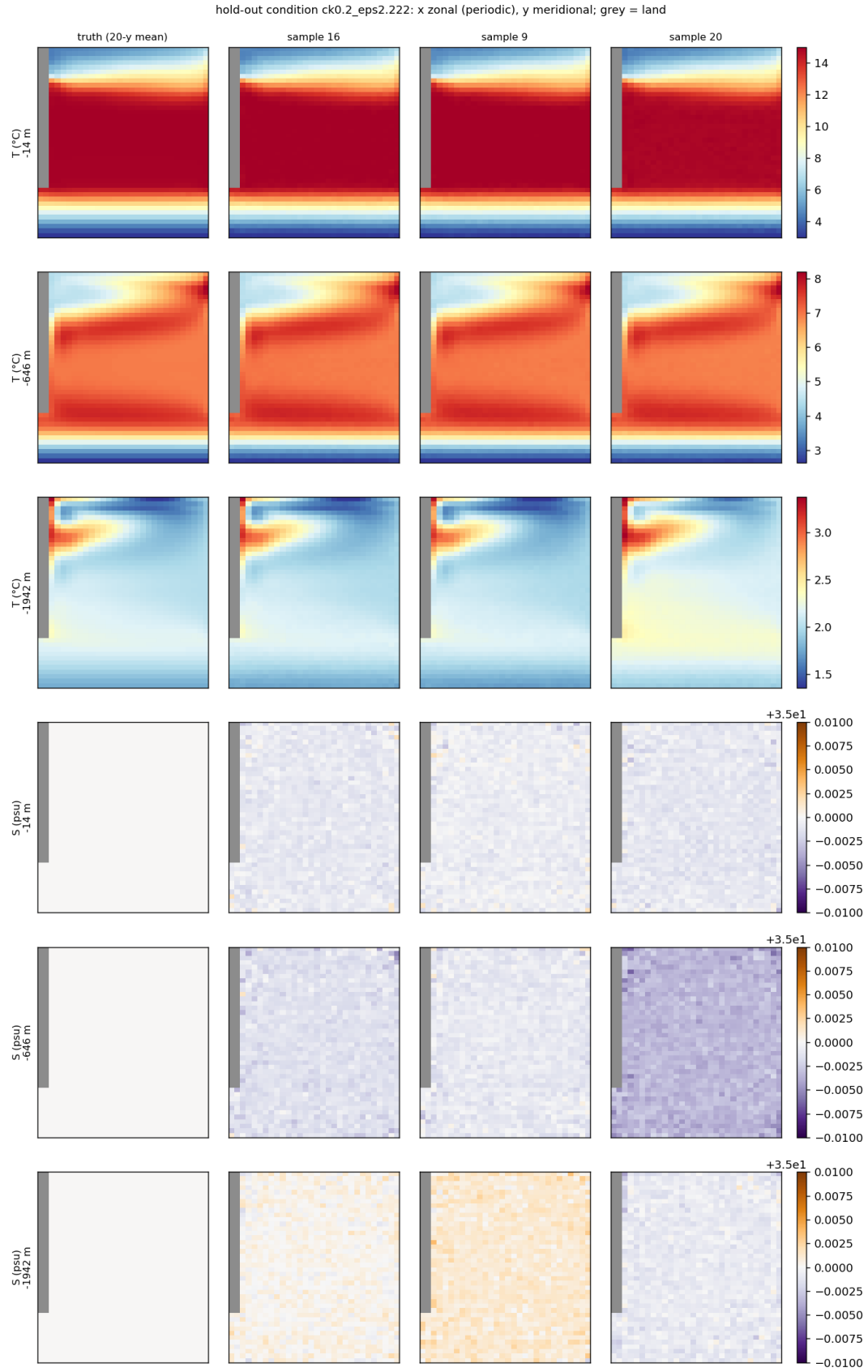


Figure 11: True 20-year mean state and three random samples of T and S at the surface, 646 m and the bottom for one held-out condition of the scattered split (grey: land; x zonal and periodic, y meridional).

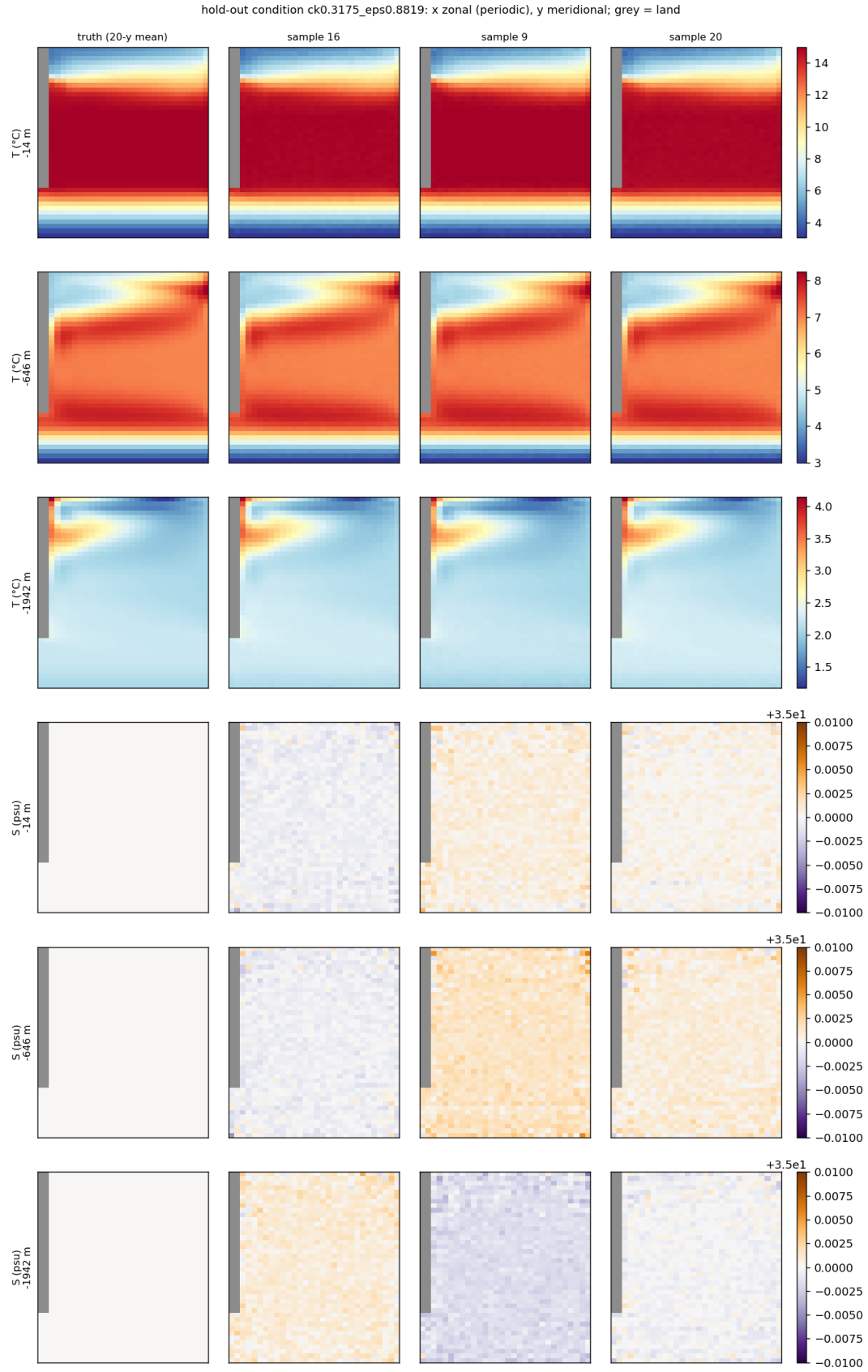


Figure 12: True 20-year mean state and three random samples of T and S at the surface, 646 m and the bottom for one held-out condition of the band split (grey: land; x zonal and periodic, y meridional).

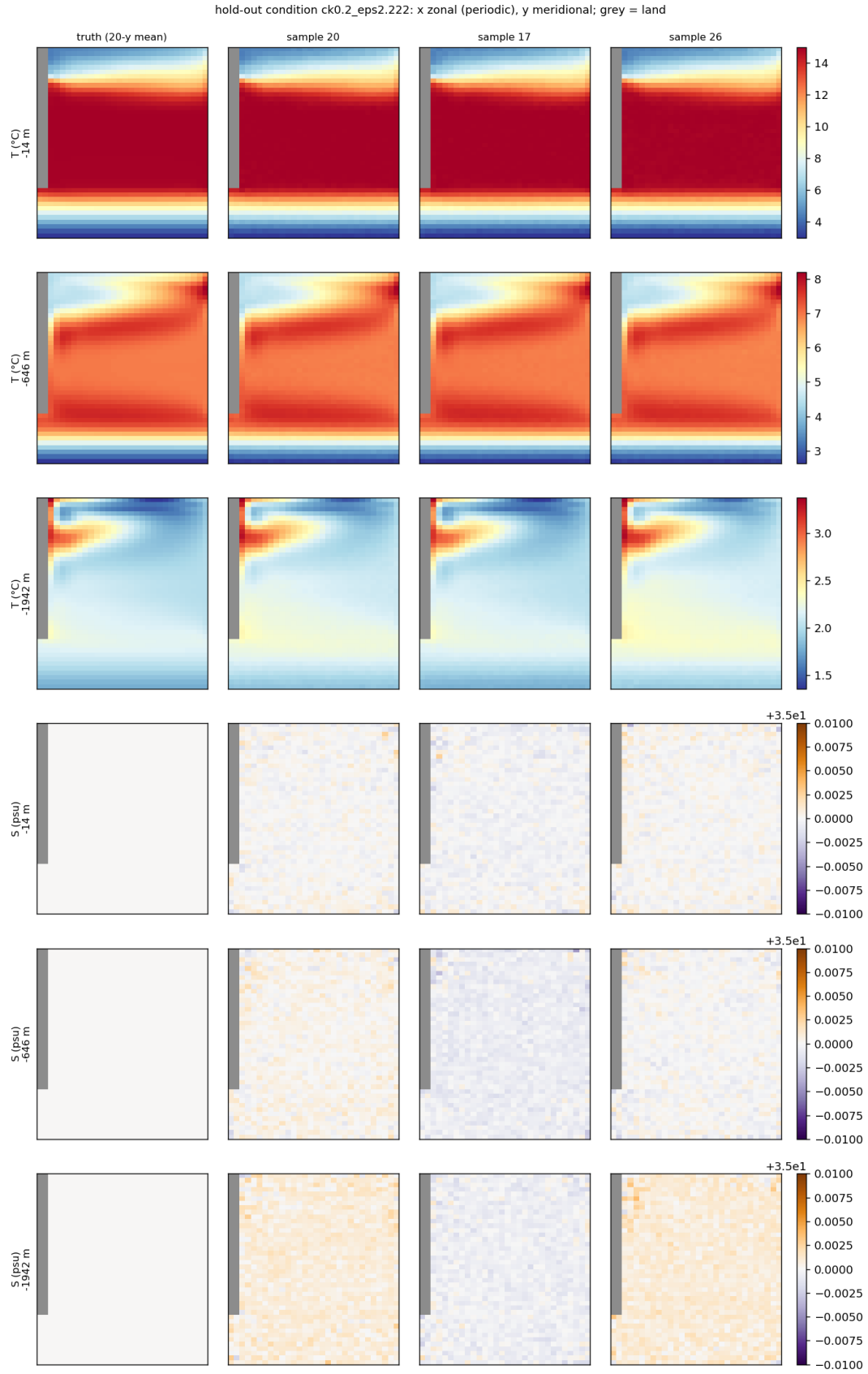


Figure 13: True 20-year mean state and three random samples of T and S at the surface, 646 m and the bottom for one held-out condition of the centre 5×5 split (grey: land; x zonal and periodic, y meridional).

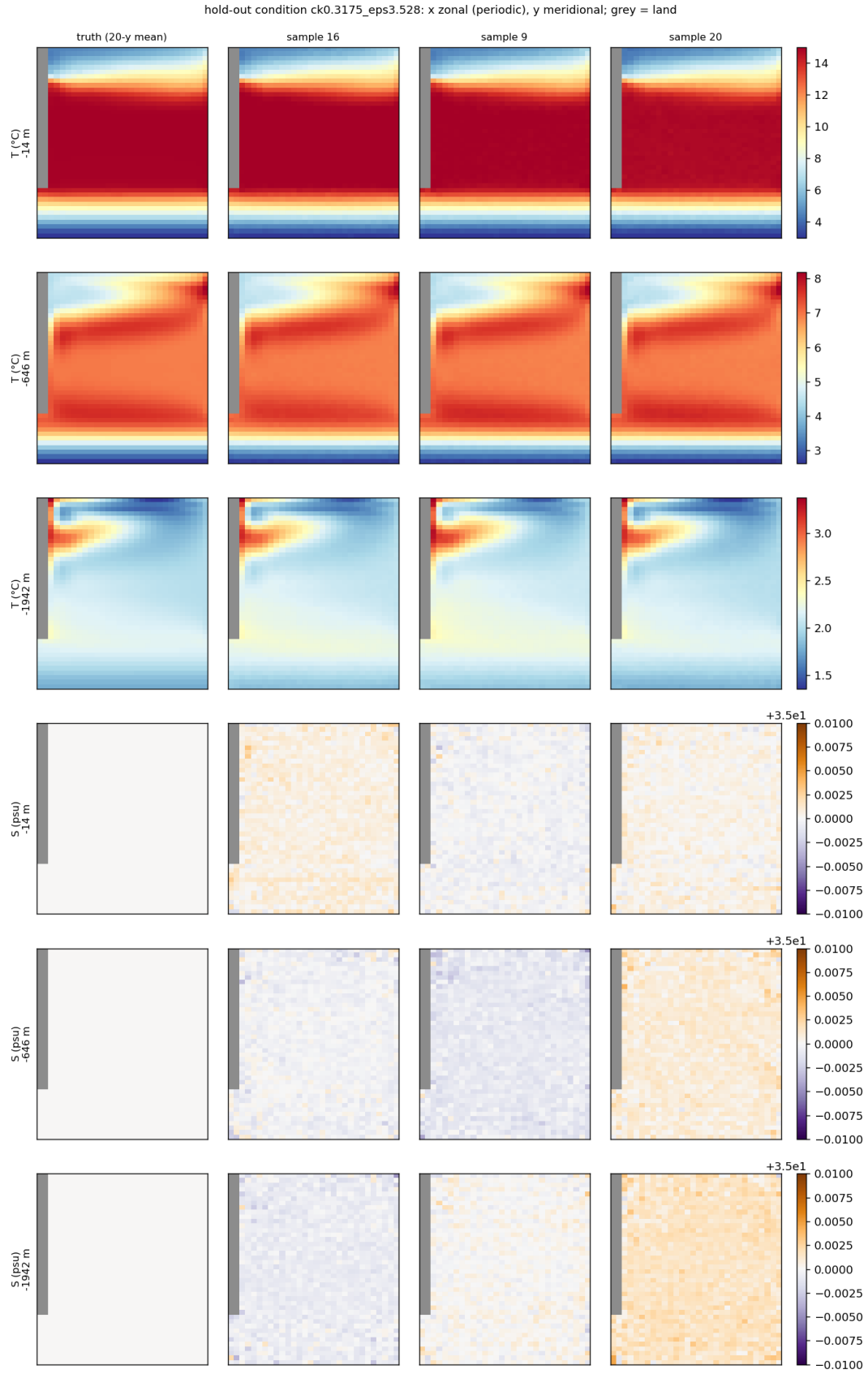


Figure 14: True 20-year mean state and three random samples of T and S at the surface, 646 m and the bottom for one held-out condition of the centre 7×7 split (grey: land; x zonal and periodic, y meridional).

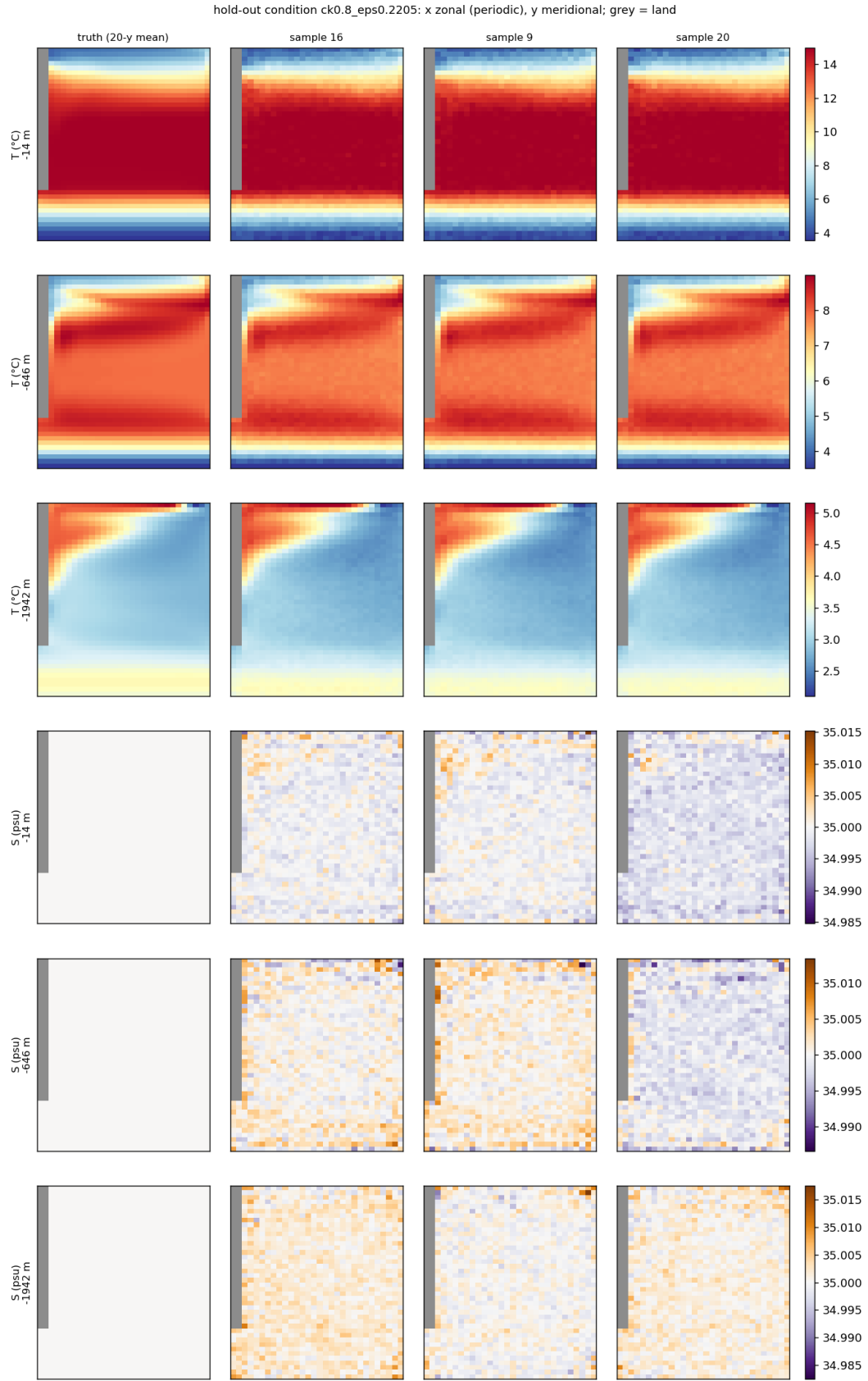


Figure 15: True 20-year mean state and three random samples of T and S at the surface, 646 m and the bottom for one held-out condition of the outer-51 split (grey: land; x zonal and periodic, y meridional).

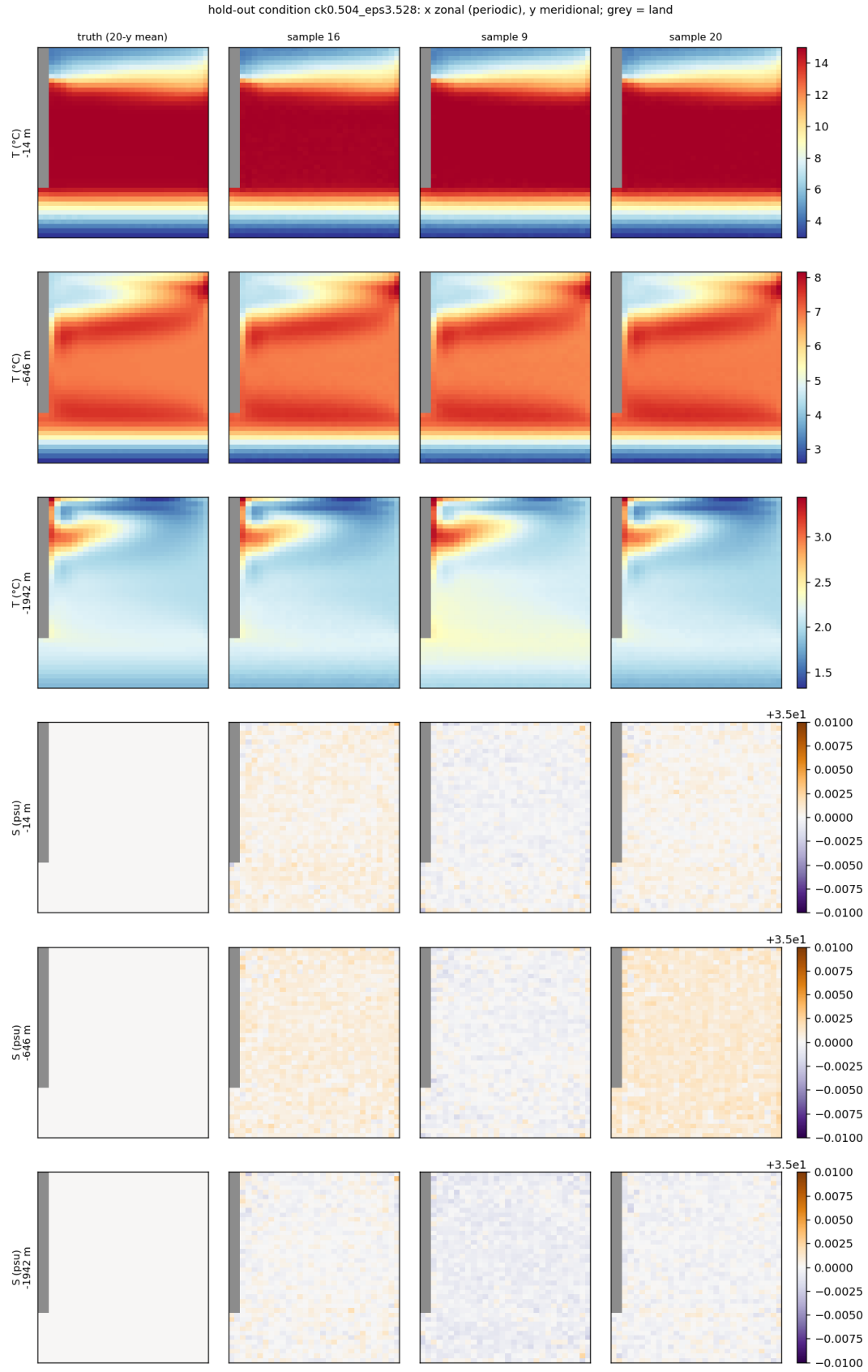


Figure 16: True 20-year mean state and three random samples of T and S at the surface, 646 m and the bottom for one held-out condition of the outer-75 split (grey: land; x zonal and periodic, y meridional).

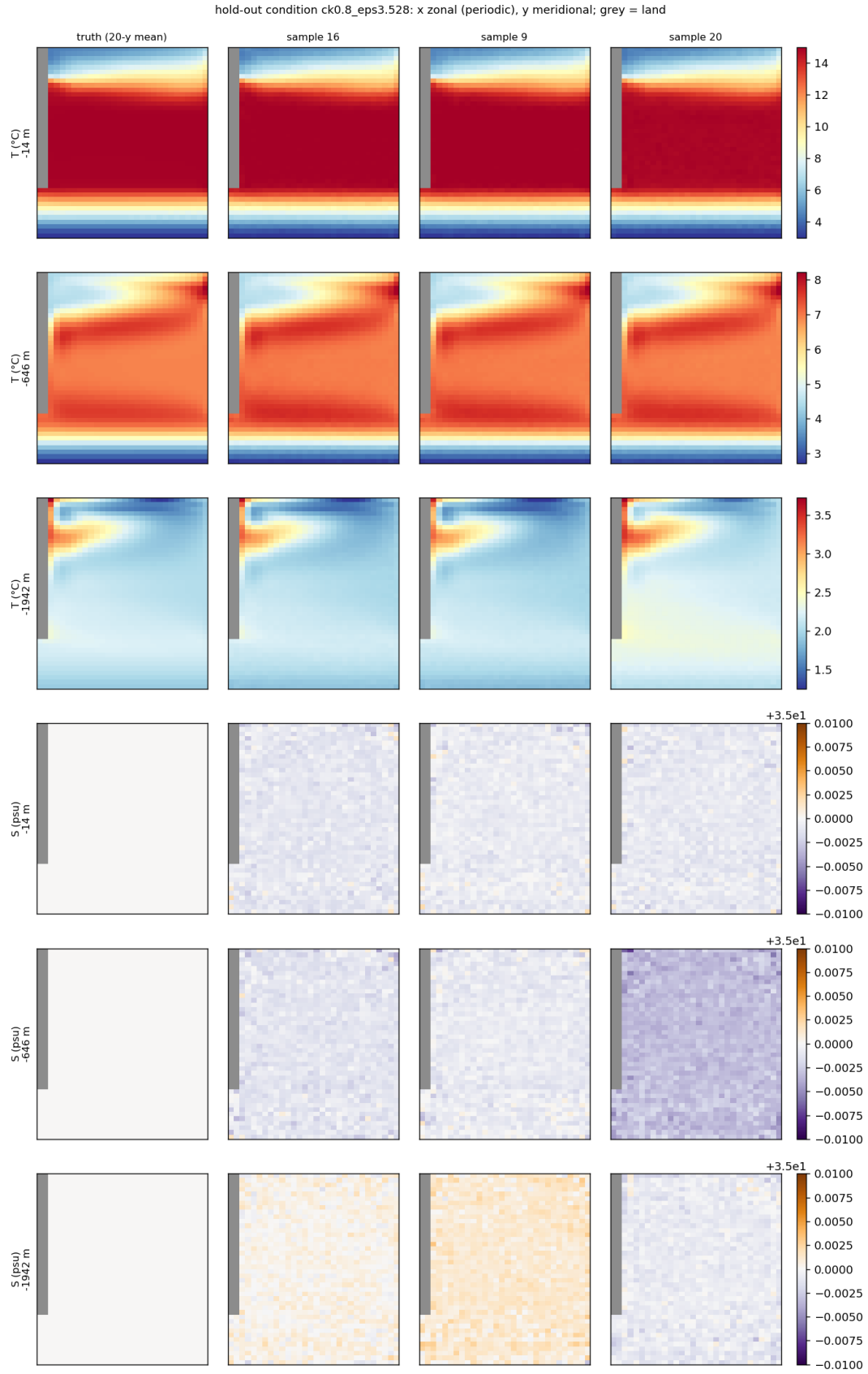


Figure 17: True 20-year mean state and three random samples of T and S at the surface, 646 m and the bottom for one held-out condition of the top-row split (grey: land; x zonal and periodic, y meridional).

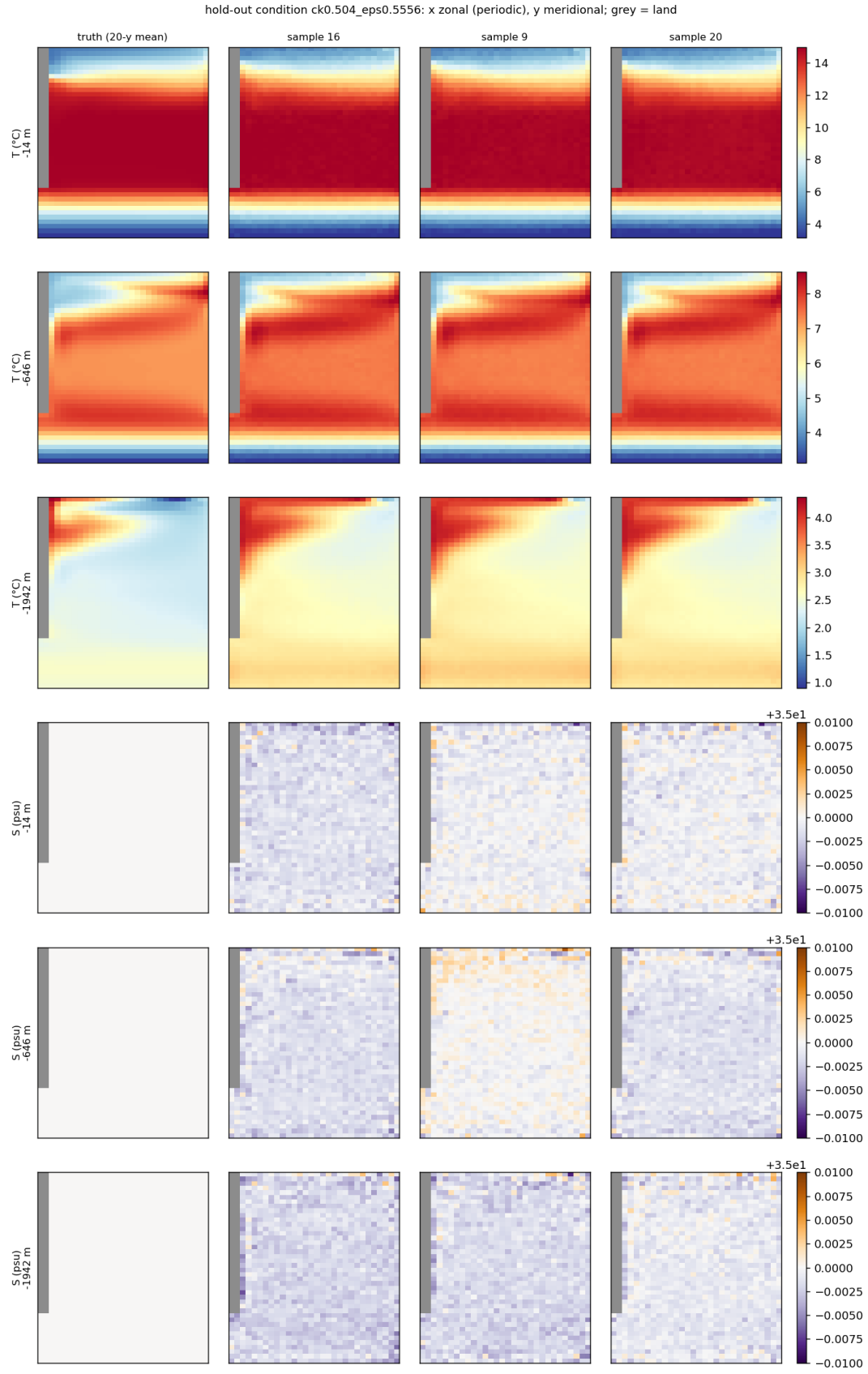


Figure 18: True 20-year mean state and three random samples of T and S at the surface, 646 m and the bottom for one held-out condition of the four-run split (grey: land; x zonal and periodic, y meridional).

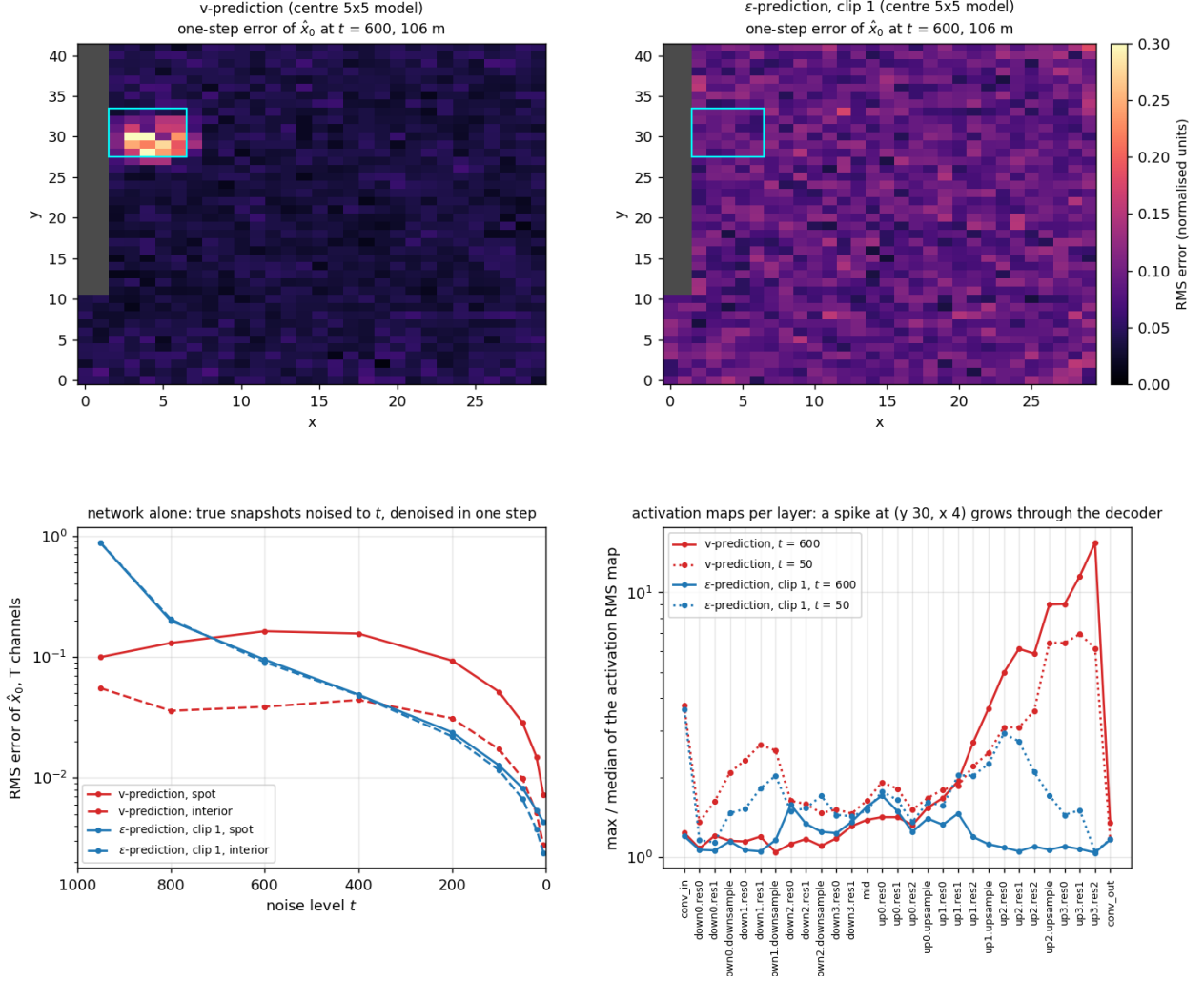


Figure 19: The decoder spike of a velocity-target model trained with the loss on all cells (centre 5×5 split). Top: RMS error of the one-step clean-state estimate for eight true snapshots of a held-out run noised to $t = 600$, at 106 m, for that network (left) and for an ϵ -prediction network (right); cyan box: the patch. Bottom left: the same error in the patch and in an interior box as a function of t . Bottom right: maximum over median of each layer's activation map: the spike appears in the second decoder block and grows through the decoder. With the loss on the water cells the maps are flat and the patch is gone.